1. Introduction to YOLO

YOLO—short for *You Only Look Once*—is a state-of-the-art real-time object detection algorithm widely used in computer vision. Unlike traditional object detection systems that scan an image multiple times, YOLO processes the entire image *once* and predicts object locations and classes in a single, unified step. This makes it extremely fast and efficient, capable of running in real time on both GPUs and small edge devices.

Object detection is the computer vision task of identifying *what* objects appear in an image and *where* they are located through bounding boxes. YOLO is a **one-stage detector**, meaning it predicts bounding boxes and class probabilities in a single forward pass of the neural network. This contrasts with two-stage detectors like R-CNN, which first generate region proposals then classify them.

2. How YOLO Works (Conceptual Breakdown)

2.1 Grid-based Prediction

YOLO divides the input image into an **S × S grid**.
Each grid cell is responsible for detecting objects whose centers fall inside the cell.

For each cell, YOLO predicts:

1. **Bounding box coordinates** (x, y, width, height)
2. **Objectness score** (confidence that an object exists)
3. **Class probabilities** (e.g., person, dog, car, etc.)

All predictions across the grid are combined to form final detections.

2.2 Bounding Box Prediction

Each predicted bounding box has:

- **x, y:** coordinates relative to the grid cell
- **w, h:** normalized width and height
- **Confidence score:** probability that a box contains an object × IoU (Intersection over Union)

YOLO uses **anchor boxes**, which are predefined box shapes, to help the model predict realistic sizes of objects.

2.3 Non-Max Suppression (NMS)

YOLO often generates multiple overlapping bounding boxes for the same object.

NMS selects the box with the highest confidence score and removes overlapping boxes based on IoU thresholds.

This ensures that only the best detection remains.

2.4 Why YOLO Is Popular

- Very fast (real-time detection)
- End-to-end single neural network
- Highly scalable (works on drones, phones, CCTV systems)
- Versatile: can detect multiple classes in various environments
- Modern versions (YOLOv5, YOLOv7, YOLOv8, YOLOv10) simplify training and deployment

Version	**Key Features**
YOLOv3	Stable, widely adopted
YOLOv4	Improved accuracy & speed
YOLOv5	PyTorch-based, popular for research
YOLOv7	Highest accuracy in 2022
YOLOv8	SOTA speed + accuracy, easiest for beginners
YOLOv10	Most efficient model (2024)

3.0 Mathematical Formulation of YOLO

3.1 Image Grid

Given an image of size $W \times H$, YOLO divides it into:

$$S \times S \text{ grid}$$

Where **S** is typically 7 (YOLOv1), 13 or 19 for later versions.

3.2 Bounding Box Prediction

Each grid cell predicts **B** bounding boxes.
A bounding box is represented mathematically as:

$$(x, y, w, h, C)$$

where

- x, y= center of bounding box (relative to cell)
- w, h= width and height (relative to whole image)
- C= confidence score

Confidence score is given by:

$$C = P(\text{object}) \times \text{IoU}$$

Where

- $P(\text{object})$= probability there is an object in that cell
- **IoU** = Intersection over Union (overlap between predicted and ground-truth)

3.3 Intersection over Union (IoU)

YOLO uses IoU to decide which prediction is best.

$$IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$